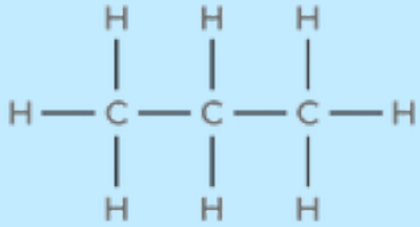
01

PROPANE

1. Molecular Formula

C_3H_8

2. Lewis Structure



3. Condensed Structural Formula

$CH_3CH_2CH_3$

4. Bond-Line Formula



- [x] atom count verified
- [x] valence checked
- [x] bond order checked
- [x] same connectivity confirmed

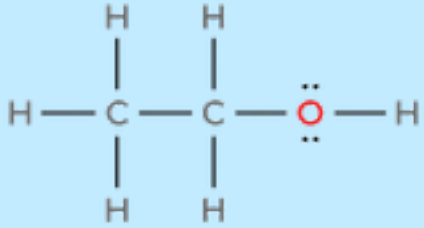
02

Ethanol

1. Molecular Formula

C_2H_6O

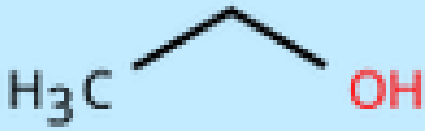
2. Lewis Structure



3. Condensed Structural Formula

CH_3CH_2OH

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

03

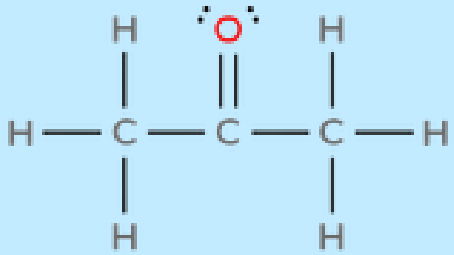
Acetone

(propan-2-one)

1. Molecular Formula

C_3H_6O

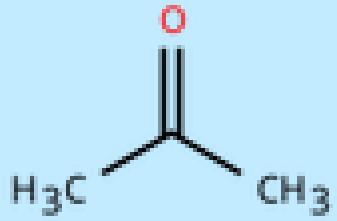
2. Lewis Structure



3. Condensed Structural Formula

$CH_3C(=O)CH_3$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

04

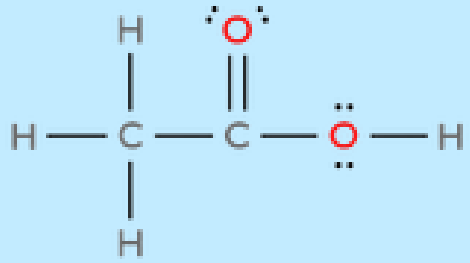
Acetic acid

(ethanoic acid)

1. Molecular Formula

$C_2H_4O_2$

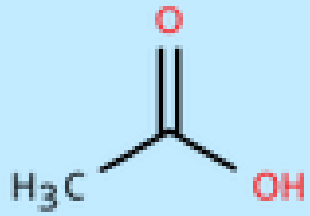
2. Lewis Structure



3. Condensed Structural Formula

$CH_3C(=O)OH$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

05

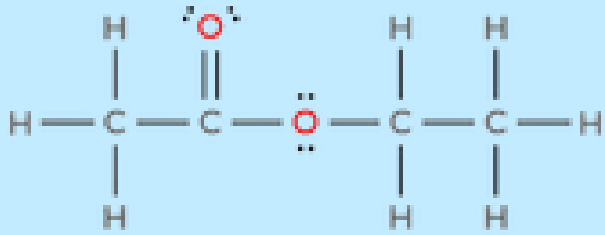
Ethyl acetate

(ethyl ethanoate)

1. Molecular Formula

$C_4H_8O_2$

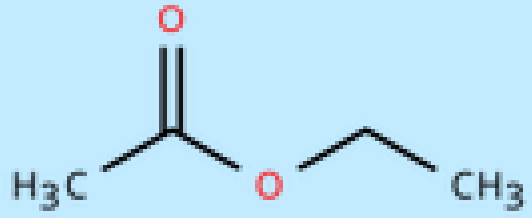
2. Lewis Structure



3. Condensed Structural Formula

$CH_3C(=O)OCH_2CH_3$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

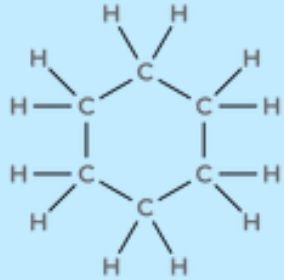
06

Cyclohexane

1. Molecular Formula

C_6H_{12}

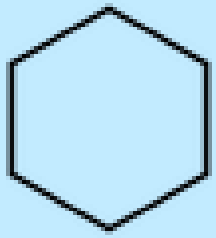
2. Lewis Structure



3. Condensed Structural Formula

cyclo-$(CH_2)_6$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

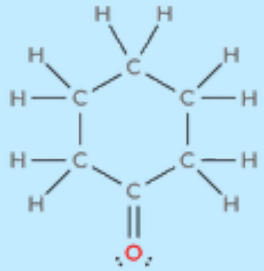
07

Cyclohexanone

1. Molecular Formula

$C_6H_{10}O$

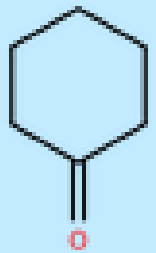
2. Lewis Structure



3. Condensed Structural Formula

cyclo-$(CH_2)_5C{=}O$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

08

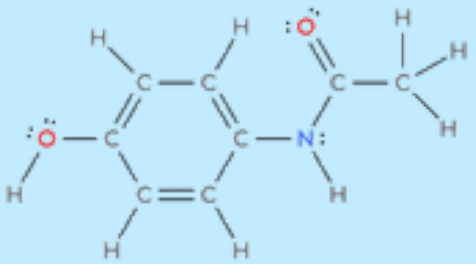
Acetaminophen

(paracetamol)

1. Molecular Formula

$C_8H_9NO_2$

2. Lewis Structure

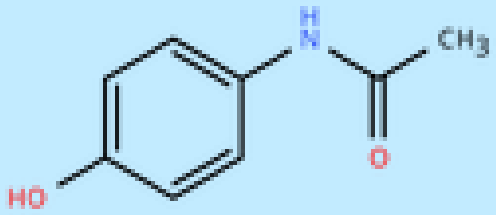


3. Condensed Structural Formula

$CH_3C(=O)NH–C_6H_4–OH$

(para, 1,4)

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

09

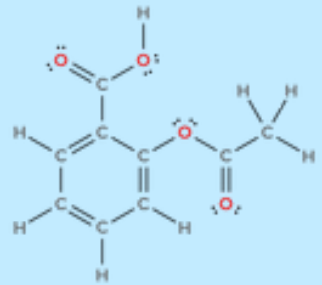
Aspirin

(acetylsalicyclic acid)

1. Molecular Formula

$C_9H_8O_4$

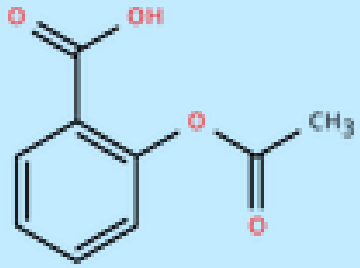
2. Lewis Structure



3. Condensed Structural Formula

$CH_3C(=O)O–C_6H_4–COOH$
(ortho, 1,2)

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

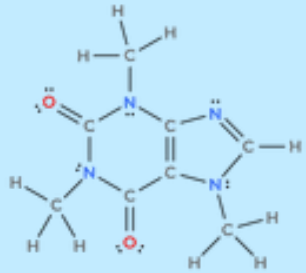
10

Caffeine

1. Molecular Formula

$$C_8H_{10}N_4O_2$$

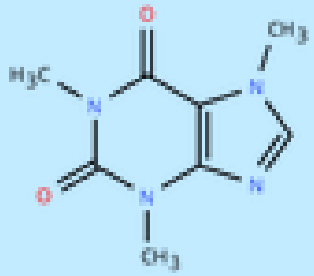
2. Lewis Structure



3. Condensed Structural Formula

$$C_5H(CH_3)_3N_4O_2$$

4. Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed